PARMAR RAILWAY

COMPLETE GK PYQ BATCH

GENERAL KNOWLEDGE

(GEOGRAPHY)

LECTURE-6

(BY- PARMAR SIR)



PHYSICAL GEOGRAPHY | INDIA GEOGRAPHY | WORLD GEOGRAPHY | ENVIRONMENT GEOGRAPHY | MAPPING CONCEPTS

Q1. Which one of the following represents the correct formula for crude birth rate (CBR), where B = live births during the year; P= Estimated mid year population of the area?

NTPC UG CBT-01 2025-26 (20 Jun, 2026 (9:00 AM - 10:30 AM)

A. (Births + Population) × 100
B. (Births × Population) ÷ 1000
C. (Population ÷ Births) × 100
D. (Births ÷ Population) × 1000

- **Crude Birth Rate (CBR):** Number of live births per 1,000 population.
  - **Formula:** (Births / Population) * 1000.

- **Data from 2011 Census of India:**
  - **Crude Birth Rate:** 21.8 per 1000 population.
  - **Crude Death Rate:** 7.1 per 1000 population.
  - **Infant Mortality Rate (IMR):** 44 per 1000 live births.
  - **Under-5 Mortality Rate:** 55 per 1000 live births.

Q2. Which of the following is NOT a pull factor of migration?

NTPC UG CBT-01 2025-26 (20 Jun, 2026 (9:00 AM - 10:30 AM)

A. Socio-economic backwardness
B. Better job opportunities
C. Peace and stability
D. Security of life

- **Migration is driven by two factors:** Pull Factors and Push Factors.

- **Pull Factors (Attract people):** Better job opportunities, better living conditions, security of life, peace, and stability.
- **Push Factors (Force people to leave):** Lack of jobs, violence, protests, poor living conditions, socio-economic backwardness.
  - Since **socio-economic** backwardness forces people to leave, it is a Push Factor, not a Pull Factor.

Q3. Which of the following densities measures the pressure of population on net cultivated area in India?

NTPC UG CBT-01 2025-26 (20 Jun, 2026 (4:30 PM - 6:00 PM)

A. Physiological density
B. Agricultural density
C. Nutritional density
D. Economic density

- **Physiological Density:** Formula is (Total Population / Net Cultivated Area). It measures the burden of the population on the available agricultural land.
- **Population Density:** Formula is (Total Population / Total Area). It means how many people live per square kilometer.

Q4. Which of the following best explains the emergence of "megaregions" with extremely high population densities (e.g., Bos Wash, Tokyo-Osaka, Pearl River Delta)?

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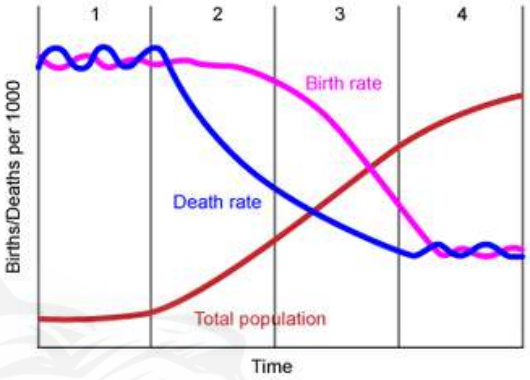
A. Megaregions form primarily because of uniform physical geography
B. They develop due to integrated economic corridors and high connectivity
C. Governments intentionally relocate populations to form megaregions
D. They exist because of high natural population growth in these regions

- **Mega-regions (like Dubai in the UAE)** develop rapidly from being barren areas to highly dense areas primarily due to the development of ports, airports, high connectivity, and economic corridors.

Q5. A country with a high birth rate and a high death rate typically reflects:

NTPC UG CBT-01 2025-26 (19 Jun, 2026 (9:00 AM - 10:30 AM)

A. Post-industrial society
B. Early stage of demographic transition
C. Low dependency ratio
D. Advanced economy



- **Demographic Transition Model - DTM:**
  - **Stage 1 (Early Stage):** Both Birth Rate and Death Rate are high. Population remains low and stable.
  - **Stage 2: Birth Rate remains high,** but Death Rate drops significantly (due to better medical facilities). This stage experiences a Population Explosion (massive population growth).
  - **Stage 3: Death Rate is low,** and Birth Rate starts to decline (due to education, female literacy, high cost of living, concepts like DINK - Double Income No Kids). Population still grows but at a slower rate. (India is entering this stage, China is fully in it).
  - **Stage 4 (Final Stage):** Both Birth Rate and Death Rate are LOW. Population stabilizes again.
  - **Stage 5 (Theoretical/Aging):** Birth rate falls below death rate, population declines (e.g., Japan, aging population).

Q6. Which of the following geographical factors most directly influences the location and growth of human settlements?

NTPC UG CBT-01 2025-26 (18 Jun, 2026 (12:45 PM - 2:15 PM)

A. Distance from the equator
B. Color of local soil
C. Availability of fresh water
D. Shape of nearby hills

- **People settle where basic survival needs are met.** Fresh water and fertile soil (like in the UP/Bihar plains around the Ganga river) attract huge populations. Factors like soil color or distance from the equator do not directly dictate human settlement.

Q7. Which one of the following combinations contributes to low population density?

NTPC UG CBT-01 2025-26 (17 Jun, 2026 (4:30 PM - 6:00 PM)

A. River valleys, floodplains, health clinics
B. Harsh climate, poor soil, few roads
C. Mild winters, abundant water, schools
D. Flat terrain, fertile soil, urban jobs

- Places with flat terrain, fertile soil, and urban jobs (like Noida) **attract high population density.**
- Places with a harsh climate, poor soil, and poor connectivity naturally **repel large populations.**

Q8. What demographic shift characterizes Brazil's transition since the 1980s?

NTPC UG CBT-01 2025-26 (18 Jun, 2026 (9:00 AM - 10:30 AM)

A. High births with low longevity
B. Rising urban mortality
C. Falling birth and death rates
D. Stable fertility with immigration

- **Brazil is shifting from a youth-heavy population** to an aging population, which is a hallmark of declining birth and death rates.

Q9. Which of the following is the most important port-based industrial region in India?

NTPC UG CBT-01 2025-26 (14 Jun, 2026 (12:45 PM - 2:15 PM)

A. Chotanagpur plateau region
B. Hooghly industrial belt
C. Hyderabad–Bengaluru corridor
D. Ahmedabad–Vadodara region

Hooghly is the only option here featuring a major port (Kolkata Port).

- **8 Major Industrial Regions in India:**
  - **Mumbai-Pune:** Famous for cotton, petroleum, engineering goods.
  - **Bengaluru-Tamil Nadu:** Famous for IT, electronics, cotton, silk.
  - **Gujarat:** Famous for diamond processing, cotton, fertilizers, petrochemicals.
  - **Chota Nagpur:** Famous for minerals (coal, iron).
  - **Visakhapatnam-Guntur:** Famous for shipbuilding, petroleum refining.
  - **Gurugram-Delhi-Meerut:** Industrial belt.
  - **Kollam-Thiruvananthapuram:** Famous for IT and agricultural processing.
  - **Hooghly:** Port-based industrial region.

Q10. According to the Census 2011, which of the following states has the highest population density among the northern Indian states?

NTPC UG CBT-01 2025-26 (14 Jun, 2026 (4:30 PM - 6:00 PM)

A. Himachal Pradesh
B. Uttarakhand
C. Rajasthan
D. Bihar

- **Bihar has the highest population density in India** among states (1,106 persons/sq km). West Bengal is second (1,028 persons/sq km).

- **Delhi has the highest population density among UTs** (11,320 persons/sq km).
- **Lowest population density state:** Arunachal Pradesh (17 persons/sq km).

Q11. Which of the following Union Territories had the lowest density of population according to Census 2011?

NTPC UG CBT-01 2025-26 (08 May, 2026 (9:00 AM - 10:30 AM)

A. Lakshadweep
B. Andaman and Nicobar Islands
C. Puducherry
D. Chandigarh

- **Andaman and Nicobar Islands have the lowest density** (46 persons/sq km).

- **Lakshadweep has the highest literacy** rate among UTs (91.9%).
- **Puducherry has the highest sex** ratio among UTs (1,037).

Q12. Which of the following countries is the youngest country in terms of the lowest median age for the year 2024?

NTPC Graduation CBT-01 2025-26 (22 Mar, 2026 Shift 1)

A. Niger
B. Palestine
C. Afghanistan
D. Iraq

- **Niger (a Central African country)** has a median age of around 15 years, meaning it has a massive youth/child population. This forms a broad-based population pyramid (indicating a future population explosion).

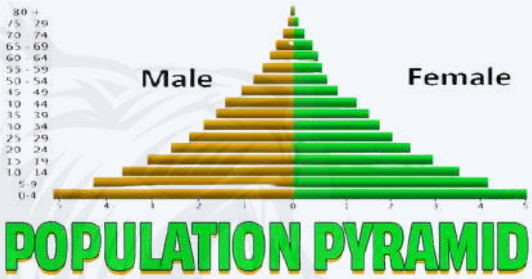
Q13. According to the UN World Population Prospects (2019), which region is projected to contribute most significantly to global population growth by 2050?

NTPC Graduation CBT-01 2025-26 (19 Mar, 2026 Shift 1)

A. Eastern Southeastern Asia
B. Europe North America
C. Sub Saharan Africa
D. Latin America and the Caribbean

- **Countries like Nigeria, Kenya, Ethiopia, and DR Congo (Sub-Saharan Africa)** have high fertility rates and broad-based population pyramids.

- **Population Pyramids:**
  - **Broad Base:** Lots of children, population will explode.
  - **Rectangular Shape:** Stable population.
  - **Tapered/Narrow Base (Bell-shaped):** Fewer children, population will decline (e.g., Japan, which has high life expectancy but low birth rate).



Q14. Which of the following states is known for the tribal settlement of the Gond community?

RRB NTPC UG CBT-01 2024 (09 Sept, 2025 Shift 1)

A. Kerala
B. Haryana
C. Punjab
D. Chhattisgarh

- **Largest Tribal Groups in India:**
  - **Bhil**: 1st largest. Found in MP, Gujarat, Chhattisgarh, Rajasthan.
  - **Gond**: 2nd largest. Found prominently in Chhattisgarh/Central India.
  - **Santhal**: 3rd largest.

Q15. Which type of rural settlement is most common in the northeastern states and Northern Plains of India?

RRB NTPC UG CBT-01 2024 (22 Aug, 2025 Shift 3)

A. Dispersed
B. Semi-clustered
C. Hamleted
D. Clustered

- **Dispersed**: Houses are spread far apart (common in hilly/forested areas like Himachal Pradesh, Uttarakhand).

- **Hamletted (Purwa/Tola):** A village divided into several distinct clusters/groups of houses.
- **Clustered:** Houses are built very close together (common in highly populated fertile plains like UP, Bihar, West Bengal).

Q16. Which physical factor most influences the formation of dispersed settlements in hilly and forested regions of India?

RRB NTPC UG CBT-01 2024 (14 Aug, 2025 Shift 1)

A. Industrial development
B. Language diversity
C. Railway networks
D. Altitude and terrain

Q17. Why can Gross Enrollment Ratio (GER) sometimes exceed 100%?

RRB NTPC UG CBT-01 2024 (07 Aug, 2025 Shift 3)

A. Due to school mergers in rural areas
B. Due to inclusion of post-graduate students
C. Due to enrollment of learners outside the official age group
D. Because of errors in census data

- **GER is calculated as** (Total Enrollment in a specific level of education, regardless of age) / (Population of the official age group for that level).
  - **It exceeds 100% because** the numerator includes students who are older or younger than the official age group (e.g., adult learners, late starters), while the denominator only counts the exact official age group.

Q18. According to the Census of India, what was the approximate decadal growth rate of India's population during 2001–2011?

RRB NTPC Graduation CBT-01 2024 (21 Jun, 2025 Shift 3)

A. 14.9%
B. 21.3%
C. 25.6%
D. 17.6%

- **Decadal Growth Rate:** 17.6%
- **Annual Growth Rate:** 1.64%
- **Highest decadal growth (State):** Meghalaya (~28%).
- **Highest decadal growth (UT):** Dadra & Nagar Haveli.
- **Sex Ratio:** India average is 943.
  - **Highest State**: Kerala (1084). **Lowest State:** Haryana (879). **Highest UT:** Puducherry (1037). **Lowest UT:** Daman & Diu.

Q19. Which of the following urban agglomerations has the highest population in India as per Census 2011?

RRB NTPC Graduation CBT-01 2024 (20 Jun, 2025 Shift 1)

A. Delhi
B. Kolkata
C. Mumbai
D. Bengaluru

- **Delhi has the highest population density.**
- Mumbai has the highest absolute population among urban agglomerations, followed by Delhi.

Q20. In which state of India is the Bhil tribal community found in large numbers?

RRB NTPC Graduation CBT-01 2024 (21 Jun, 2025 Shift 2)

A. Odisha
B. Tamil Nadu
C. Madhya Pradesh
D. Kerala

Q21. Kanpur, a city in the Indian state of Uttar Pradesh, is an example of which type of town?

RRB NTPC Graduation CBT-01 2024 (19 Jun, 2025 Shift 2)

A. Administrative Town
B. Recreational Town
C. Industrial Town
D. Transfer and Distribution Town

- **Kanpur is an industrial town famous** for its Textile and Leather industries.

Q22. In the Indian context, population growth in Class I cities is the most influenced by _____.

RRB NTPC Graduation CBT-01 2024 (18 Jun, 2025 Shift 1)

A. Natural increase
B. migration from rural areas
C. religious factors
D. government employment

- **Class I Cities:** Cities with a population of more than 1 Lakh (100,000).
- **Population change is driven by Births, Deaths, and Migration** (In-migration & Out-migration). Urban growth in Class I cities (like Delhi, Mumbai) is heavily driven by rural-to-urban migration for jobs.

Q23. During which phase and year was a steady population growth rate observed in India?

RRB NTPC Graduation CBT-01 2024 (12 Jun, 2025 Shift 3)

A. Phase III: 1951-1981
B. Phase IV: 1981-2011
C. Phase I: 1891-1921
D. Phase II: 1921-1951

- **Phase 1 (1901-1921):** Stagnant/Stationary population (High birth and high death rates). 1921 is known as the "Year of Great Divide" because population started growing consistently after this.
- **Phase 2 (1921-1951):** Steady population growth.
- **Phase 3 (1951-1981):** Population Explosion (Sharp decline in mortality, persistently high fertility).
- **Phase 4 (1981-Present):** Stable growth showing significant signs of slowing down.

Q24. The Konyak tribe of India is mainly found in which of the following states of India?

RRB NTPC Graduation CBT-01 2024 (06 Jun, 2025 Shift 3)

A. Uttar Pradesh and Bihar
B. Nagaland and Arunachal Pradesh
C. Arunachal Pradesh and Uttar Pradesh
D. Nagaland and Sikkim

Q25. In India, a city with a population of less than 10,000 and more than 5000 is classified as which type of class town?

RRB NTPC Graduation CBT-01 2024 (06 Jun, 2025 Shift 2)

A. Class I
B. Class II
C. Class V
D. Class VI

- **Class I:** > 100,000
- **Class II:** 50,000 - 100,000
- **Class III:** 20,000 - 50,000
- **Class IV:** 10,000 - 20,000
- **Class V:** 5,000 - 10,000
- **Class VI:** < 5,000

Q26. Which period in India's demographic history is marked by a sharp decline in mortality and persistently high fertility rates?

RRB NTPC Graduation CBT-01 2024 (05 Jun, 2025 Shift 3)

A. Phase II (1921-1951)
B. Phase IV (1981-2001)
C. Phase I (1901-1921)
D. Phase III (1951-1981)

Q27. In a population pyramid, a narrow base and a wider top are most likely to indicate which of the following?

NTPC UG CBT-01 2025-26 (14 Jun, 2026 (12:45 PM - 2:15 PM)

A. High birth rates and high death rates
B. Low birth rates and low life expectancy
C. High birth rates and low death rates
D. Low birth rates and high life expectancy

- **Narrow base** = Few children (Low birth rate).
- **Wider top** = Many older people living long lives (High life expectancy, indicating a developed country like Japan).

Q28. What is the primary reason for the rapid population growth in developing countries during the second stage of the Demographic Transition Model?

NTPC UG CBT-01 2025-26 (09 May, 2026 (12:45 PM - 2:15 PM)

A. High birth rates and high death rates
B. Low birth rates and low death rates
C. High birth rates and declining death rates
D. Low birth rates and high death rates



TRANSPORT

Q29. Which of the following is a landlocked inland river port in India?

NTPC UG CBT-01 2025-26 (19 Jun, 2026 (12:45 PM - 2:15 PM)

A. Kandla port
B. Chennai port
C. Mumbai port
D. Pandu port

- **Pandu Port:** Located on the Brahmaputra River in Assam (North-East). It is an inland river port.
- **Kolkata Port:** Another major Riverine port (on the Hooghly river). Also known as Syama Prasad Mookerjee Port.
- **Kandla Port:** A Tidal port in Gujarat, renamed Deendayal Port.
- **Visakhapatnam:** The deepest landlocked sea port (but not a riverine port).
- **Kochi:** Known as the "Queen of the Arabian Sea".

Q30. Which of the following is an inland riverine port?

NTPC UG CBT-01 2025-26 (19 Jun, 2026 (12:45 PM - 2:15 PM)

A. Kochi
B. Kolkata
C. Visakhapatnam
D. Kandla

Q31. Which Indian port situated at the entrance of the Zuari estuary gained significance after remodelling in 1961 to handle iron-ore exports to Japan?

NTPC UG CBT-01 2025-26 (17 Jun, 2026 (12:45 PM - 2:15 PM)

A. Jawaharlal Nehru Port
B. New Mangalore Port
C. Kandla Port
D. Marmagao Port

- **Located in Goa**. It is India's leading port for Iron Ore export.
- **Major Public Ports in India:** There are 13 major ports in total. 12 are public, and 1 is private/corporatized (Ennore/Kamarajar in Tamil Nadu).

Q32. Most of the route of the Orient Express (Trans-Continental Railway Route) majorly lies in which continent?

NTPC UG CBT-01 2025-26 (13 Jun, 2026 (12:45 PM - 2:15 PM)

A. North America
B. Asia
C. Europe
D. South America

Created in 1883, it connects Paris (France) to Istanbul (Turkey).

Q33. TheGolden Quadrilateral connects which of the following cities?

NTPC Graduation CBT-01 2025-26 (22 Mar, 2026 Shift 2)

A. Mumbai, Pune, Hyderabad, Bangalore
B. Delhi, Bhopal, Ahmedabad, Mumbai
C. Delhi, Mumbai, Chennai, Kolkata
D. Chennai, Bangalore, Hyderabad, Kolkata



- **Golden Quadrilateral Launched in 1999** under PM Atal Bihari Vajpayee, completed in 2012. Total length: 5,846 km.
  - **Longest stretch:** Kolkata to Chennai (1,684 km) along NH-16.
  - **Shortest stretch**: Chennai to Mumbai (1,290 km) along NH-48.
  - **Longest National Highway:** NH-44 (Srinagar to Kanyakumari).
- **North-South Corridor:** Srinagar to Kanyakumari (approx 4,000 km).
- **East-West Corridor:** Silchar (Assam) to Porbandar (Gujarat) (approx 3,300 km).
  - **Both corridors intersect at Jhansi (UP).**

Q34. Which of the following terms is used for places that serve as the meeting point of two or more transport routes?

NTPC Graduation CBT-01 2025-26 (22 Mar, 2026 Shift 1)

A. Grid
B. Link
C. Network
D. Node

Q35. On which of the following National Waterways of India is the floating terminal Farakka situated?

NTPC Graduation CBT-01 2025-26 (18 Mar, 2026 Shift 2)

A. National Waterways 3
B. National Waterways 5
C. National Waterways 2
D. National Waterways 1

- **NW-1:** Ganga-Bhagirathi-Hooghly river system. From Prayagraj to Haldia. It is the longest national waterway.
  - Farakka is a barrage in West Bengal on this route. (Farakka agreement signed in 1996).

- **NW-2:** Brahmaputra River.
- **NW-3:** Kollam to Kottapuram (Kerala).
- **NW-4:** Godavari-Krishna region (Andhra Pradesh / Kakinada).
- **NW-5:** Odisha/West Bengal region.

Q36. Which international waterway is most vital for world trade, linking the Mediterranean Sea with the Red Sea?

NTPC Graduation CBT-01 2025-26 (19 Mar, 2026 Shift 3)

A. Panama Canal
B. Strait of Malacca
C. Bosporus Strait
D. Suez Canal



- **It is a man-made canal,** built between 1859-1869. Controlled primarily by Egypt, contributing significantly to its GDP.
  - **Length: 193.3 km.**

- **Important Straits nearby:**
  - **Strait of Hormuz:** Connects Persian Gulf to Gulf of Oman.
  - **Bab-el-Mandeb (Gateway of Tears):** Connects Red Sea to Gulf of Aden.
  - **Strait of Gibraltar** (Connects Mediterranean to Atlantic – shown on map).
- **Before the Suez Canal, ships had to travel** around Africa via the Cape of Good Hope (the route Vasco da Gama took).

Q37. The Trans-Siberian Railways major rail route of Russia runs from St. Petersburg in the west to _____ on the Pacific Coast in the east.

NTPC Graduation CBT-01 2025-26 (24 Mar, 2026 Shift 3)

A. Vladivostok
B. Novosibirsk
C. Moscow
D. Irkutask



- **It is the longest rail route in the world (approx 9,300 km).**
  - It crosses 8 time zones in Russia.

Q38. Why is air transport less popular for freight movement in India?

RRB NTPC UG CBT-01 2024 (28 Aug, 2025 Shift 3)

A. It is widely available in rural and remote villages.
B. It requires minimal infrastructure for operation.
C. It is expensive and ideal only for light, high-value goods.
D. It is slower and less efficient than water transport

- **Fuel costs are very high,** limiting cargo to light, expensive items.

Q39. The vast network of railways that helped people meet their family and relatives was pioneered during the latter half of which of the following centuries in India?

RRB NTPC UG CBT-01 2024 (12 Aug, 2025 Shift 2)

A. 19th
B. 17th
C. 20th
D. 18th

- The **first passenger train in India** ran between Bombay and Thane in 1853 (which falls in the 19th century) during Lord Dalhousie's time.

Q40. The Hazira-Vijaipur-Jagdishpur (HVJ) pipeline is mainly used to transport:

RRB NTPC UG CBT-01 2024 (12 Aug, 2025 Shift 3)

A. Natural gas
B. petrol
C. crude oil
D. kerosene

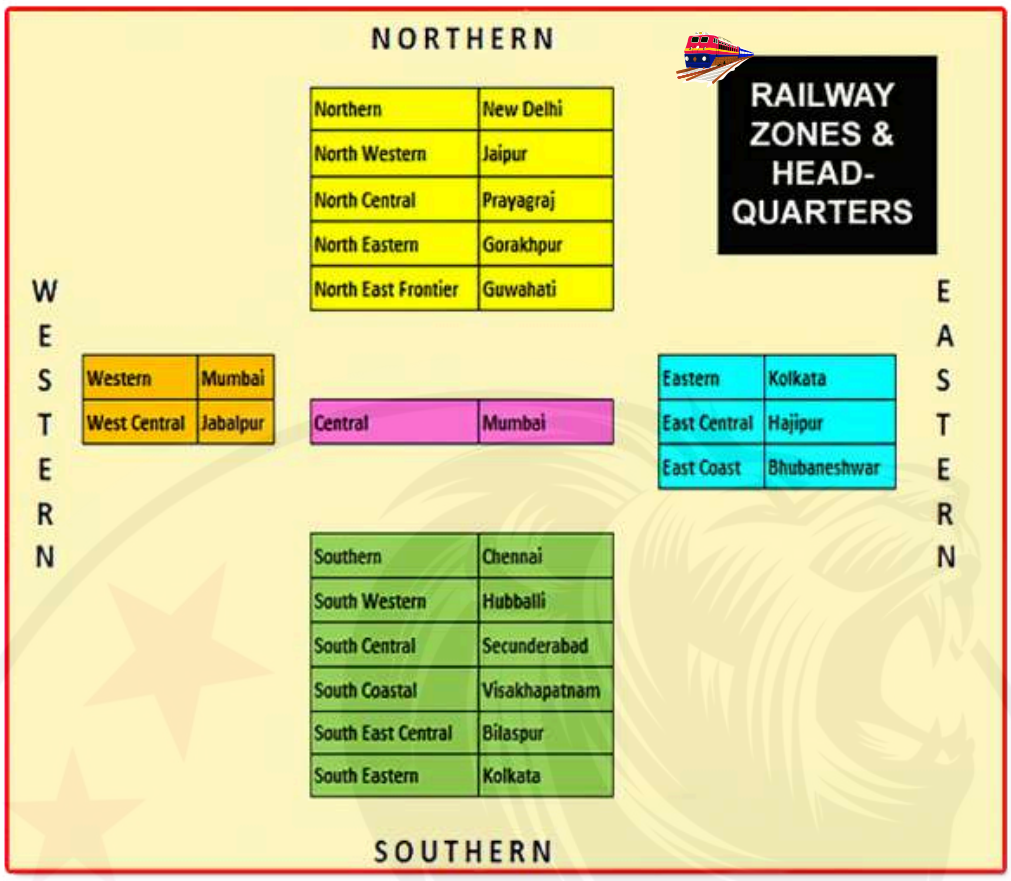
- **HVJ stands for Hajira (Gujarat) – Vijaipur (Madhya Pradesh) – Jagdishpur (Uttar Pradesh).**
  - **It is operated by GAIL** (Gas Authority of India Limited).
  - **Construction started around 1986** and was completed alongside the formation of GAIL in 1987.

Q41. Where is the headquarters of the North Western Railway zone located?

RRB NTPC UG CBT-01 2024 (07 Aug, 2025 Shift 3)

A. Jaipur
B. Hubli
C. New Delhi
D. Hajipur

- **North Western: Jaipur**



Q42. Pamban Port is located in which state of India?

RRB NTPC UG CBT-01 2024 (07 Aug, 2025 Shift 2)

A. Tamil Nadu
B. Maharashtra
C. Karnataka
D. Kerala

Q43. Which of the following is a major challenge facing the development of inland waterways in India?

RRB NTPC Graduation CBT-01 2024 (23 Jun, 2025 Shift 2)

A. Minimal environmental concerns
B. High levels of private investment
C. Lack of adequate terminal facilities
D. Well-established infrastructure

Q44. What is the major reason behind higher road density in Kerala compared to Rajasthan?

RRB NTPC Graduation CBT-01 2024 (17 Jun, 2025 Shift 2)

A. Higher Gross Domestic Products
B. Mineral distribution
C. Topography and population density
D. Availability of land

- **Rajasthan has a harsh desert topography** and low population density, reducing the need for dense road networks.
- **Kerala has a suitable topography** and very high population density requiring dense road connectivity.

MINERALS

Q45. The Damodar Valley of the Indian sub-continent is mostly rich in which mineral deposit?

NTPC UG CBT-01 2025-26 (19 Jun, 2026 (4:30 PM - 6:00 PM)

A. Coal
B. Iron
C. Magnesium
D. Limestone

- **Damodar Valley Corporation (DVC):** Established in 1948 based on the USA's Tennessee Valley Authority (TVA) model.
  - It was India's first multipurpose river valley project.

- **Jawaharlal Nehru called dams the "Temples of Modern India".**
- **4 Dams built under DVC:** Tilaiya, Konar, Maithon, Panchet.

Q46. Kudremukh in Karnataka is famous for the mining of:

NTPC UG CBT-01 2025-26 (16 Jun, 2026 (9:00 AM - 10:30 AM)

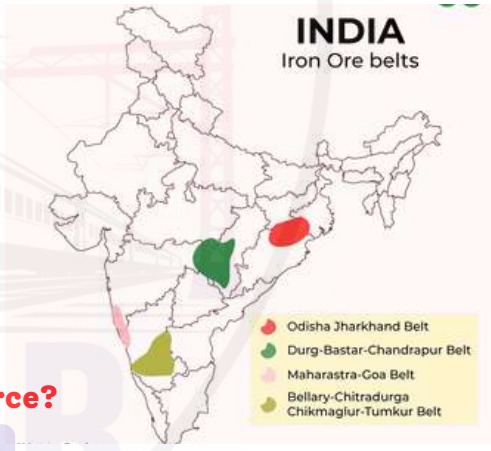
A. Iron Ore
B. Manganese
C. Limestone
D. Copper

- **"Kudre-mukh" means Horse's face/hoof.**
  - It is located in the Bellary-Chitradurga-Chikkamagaluru-Tumakuru iron ore belt in Karnataka.
  - It **predominantly mines the Magnetite variety of iron ore** (highest iron content).
- (Hematite ore is mostly found in the Odisha-Jharkhand belt).

Q47. Which region of Maharashtra is known for iron ore production?

NTPC UG CBT-01 2025-26 (08 May, 2026 (4:30 PM - 6:00 PM)

A. Chandrapur and Bhandara
B. Aurangabad and Jalna
C. Pune and Thane
D. Nagpur and Wardha



- **Located in the Maharashtra-Goa iron ore belt** (Durg-Bastar-Chandrapur belt extending into Maharashtra/Gadchiroli region).

Q48. Which of the following is a non-metallic natural resource?

A. Copper
B. Limestone
C. Gold
D. Iron ore

NTPC Graduation CBT-01 2025-26 (22 Mar, 2026 Shift 3)

- **Gold Mines in India:** KGF (Kolar Gold Fields) and Hutti Gold Mines, both in Karnataka.

Q49. Which plateau belt in India is notably rich in both iron ore and coal resources?

NTPC Graduation CBT-01 2025-26 (18 Mar, 2026 Shift 3)

A. North-Eastern Plateau Region
B. South-Western Plateau Region
C. West-Eastern Plateau Region
D. North-Western Plateau Region

Q50. Jhunjhunu and Alwar districts are known for the production of which minerals?

NTPC Graduation CBT-01 2025-26 (17 Mar, 2026 Shift 3)

A. Bauxite
B. Gold
C. Mica
D. Copper

- **The Khetri mines in Jhunjhunu (Rajasthan)** are famous for copper.
  - **Copper ore example:** Malachite.
  - **Copper was the first metal invented/used by humans** (Chalcolithic age).
- **Mica is a non-metallic mineral** with high dielectric strength.
  - It is a bad conductor of heat and electricity, making it an excellent insulator. Also used in cosmetics.
  - **Top Producer:** Andhra Pradesh (Nellore district), followed by Rajasthan.
- **Bauxite (Aluminum Ore):** Odisha is the top producer (Panchpatmali deposits).

Q51. Which factor has contributed most to the depletion of high-grade mineral ores in India over recent decades? NTPC Graduation CBT-01 2025-26 (24 Mar, 2026 Shift 1)

A. Limited foreign investment in mining
B. Overdependence on coal-based industries
C. Increased export of unprocessed minerals
D. Selective mining of rich ores

Q52. Manganese nodules are primarily found on: RRB NTPC UG CBT-01 2024 (04 Sept, 2025 Shift 3)

A. river deltas
B. ocean beds
C. glacier surfaces
D. mountain slopes

- **Samudrayaan Mission:** India's mission to explore polymetallic nodules in the deep ocean. Uses a submersible vehicle called MATSYA 6000, made of Titanium, carrying 3 humans to a depth of 6,000 meters.
  - **Developed by NIOT** (National Institute of Ocean Technology) based in Chennai.

Q53. Which of the following minerals is known for its insulating properties and is used in electrical industries? RRB NTPC UG CBT-01 2024 (14 Aug, 2025 Shift 1)

A. Bauxite
B. Limestone
C. Mica
D. Iron ore

- **Odisha, Chhattisgarh, and Jharkhand—together with Karnataka and Goa**—form the primary heartland of iron ore reserves and production in India, accounting for over 95% of the country's total deposits

- **In Odisha:** Keonjhar, Sundargarh, Mayurbhanj, and Jajpur (Barabil-Koira valley).
- **In Chhattisgarh:** Dantewada (Bailadila range, the largest mechanized mine in Asia) and Durg (Dalli-Rajhara range).
- **In Jharkhand:** West Singhbhum (Noamundi and Gua mines) and Palamu (magnetite near Daltonganj).

Q54. Which of the following minerals is known as 'Brown Gold' in India? RRB NTPC UG CBT-01 2024 (13 Aug, 2025 Shift 1)

A. Mica
B. Copper
C. Bauxite
D. Lignite

- **Coal is also known as "Buried Sunshine".** It is a non-renewable fossil fuel.
- **Indian Coal origins:** Mostly Gondwana coal (high quality). Tertiary coal (lower quality) is found in the North-East (e.g., Namchik-Namphuk in Arunachal Pradesh) and J&K.

- **Types of Coal:**
  - **Anthracite**: Highest grade, high carbon, low moisture. Found mainly in Jammu & Kashmir.
  - **Bituminous**: Most abundant in India. Used as "Metallurgical Coal" for purifying ores (smelting).
  - **Lignite**: Known as "Brown Coal". Major source is Neyveli in Tamil Nadu.
  - **Peat**: Lowest grade, very high moisture.

Q55. Which of the following is a mineral-based industry that uses ore or other minerals as raw materials?

RRB NTPC Graduation CBT-01 2024 (21 Jun, 2025 Shift 2)

A. Rubber
B. Iron and steel
C. Textile
D. Sugar

- **Basic Industry (Heavy Industry):** Industries that depend heavily on bulky raw materials (like Iron and Steel plants depending on coal and iron ore) are set up near the source of the raw materials to save on transport costs (**e.g.**, Rourkela in Odisha, Durgapur in WB, Bhilai in Chhattisgarh).
- **Footloose Industries:** Light industries not tied to raw material locations. Their raw materials and finished goods are light and easy to transport (**e.g.**, Diamond processing. Panna in MP mines diamonds, but processing happens in Surat, Gujarat).

Dams, Lakes and Waterfall

Q56. The Sardar Sarovar Project is built on which river?

NTPC UG CBT-01 2025-26 (20 Jun, 2026 (12:45 PM - 2:15 PM)

A. Mahi
B. Narmada
C. Tapi
D. Sabarmati

- **Narmada** River: Sardar Sarovar Dam and Indira Sagar Dam.
- **Tapi River:** Ukai Dam.
- **Chambal River:** Gandhi Sagar Dam (MP/Ratlam), Rana Pratap Sagar Dam (Rajasthan), Jawahar Sagar Dam (Rajasthan).
- **Mahi River:** The only river in India that cuts the Tropic of Cancer twice.

Q57. Which of the following is the largest multipurpose river valley project in India?

NTPC UG CBT-01 2025-26 (18 Jun, 2026 (12:45 PM - 2:15 PM)

A. Sardar Sarovar
B. Bhakra Nangal
C. Damodar Valley
D. Hirakud

- **Located on the Sutlej River.** It is the highest gravity dam and largest overall.
  - **Bhakra is in Himachal Pradesh; Nangal is in Punjab.**
- **Tehri Dam** (Bhagirathi river) is the highest earthen dam.
- **Hirakud** (Mahanadi) is the longest dam.

Q58. The largest freshwater lake in India is:

NTPC UG CBT-01 2025-26 (14 Jun, 2026 (12:45 PM - 2:15 PM)

A. Loktak Lake
B. Dal Lake
C. Chilika Lake
D. Wular Lake

- **Located in J&K**, fed by the Jhelum river.
- **Dal Lake:** Known as the "Jewel of Srinagar".

- **Chilika Lake:** Largest overall lake and largest brackish/saltwater lagoon in India (Odisha).
- **Loktak Lake:** Manipur. Shows a centripetal drainage pattern. Has Keibul Lamjao floating national park.
  - **Note**: **Mizoram is known as the "Molasses Basin"**

Q59. Chilika Lake spans across which three districts of Odisha?

NTPC Graduation CBT-01 2025-26 (19 Mar, 2026 Shift 2)

A. Puri, Jagatsinghpur and Khurda
B. Cuttack, Puri and Kendrapara
C. Puri, Khurda and Ganjam
D. Khurda, Balasore and Ganjam

Q60. Vembanad Lake, India's longest lake facing severe degradation, is located in which state?

RRB NTPC UG CBT-01 2024 (01 Sept, 2025 Shift 2)

A. Assam, home to the Brahmaputra river basin
B. Kerala, known for its backwaters and lagoons
C. West Bengal, near the Sundarban mangrove forests
D. Odisha, famous for Chilika Lake ecosystem

- **Kerala is famous for its backwaters,** locally known as Kayals.
- **Punnamada Kayal:** Hosts the famous Nehru Trophy Boat Race during the Onam festival.
- **Lagoon Lakes:** Found on the coast. **Examples**: Chilika (Odisha), Vembanad (Kerala), Pulicat (border of Andhra and Tamil Nadu).
- **Kolleru Lake**: An inland freshwater lake between the Krishna and Godavari deltas in Andhra Pradesh (NOT a lagoon).

Q61. The Koyna Project is constructed on a tributary of the River Krishna in which state?

RRB NTPC UG CBT-01 2024 (28 Aug, 2025 Shift 2)

A. Karnataka
B. Tamil Nadu
C. Maharashtra
D. Andhra Pradesh

- **Koyna is a tributary of Krishna River** (Maharashtra).
- **Lonar Lake:** A crater lake in Buldhana district, Maharashtra (formed by meteor impact or volcanic crater).

- **Pangong Tso and Tso Moriri:** Both are saltwater lakes in Ladakh.
- **Highest Lake in India:** Guru Dongmar Lake in Sikkim.

Q62. One of the major natural lakes in India, the Sambhar Lake of Rajasthan, is located in which of the following districts of the state?

RRB NTPC UG CBT-01 2024 (19 Aug, 2025 Shift 3)

A. Jaipur
B. Jhunjhunu
C. Barmer
D. Jaisalmer

- **Sambhar is the Largest Inland Saltwater Lake in India.** (Chilika is the largest overall but is a coastal lagoon).
- **Pushkar Lake and Ana Sagar Lake** are located in Ajmer, Rajasthan.
- **Largest Oxbow Lake in Asia:** Kanwar Lake (Kabar Taal) in Bihar.
- **Montreux Record:** A register of wetland sites under the Ramsar Convention facing severe ecological threats**. India has two:** Loktak Lake (Manipur) and Keoladeo Ghana National Park (Rajasthan).

Q63. Which of the following projects is designed to provide irrigation to Rajasthan's desert areas?

RRB NTPC UG CBT-01 2024 (13 Aug, 2025 Shift 3)

A. Indira Gandhi Canal
B. Hirakud Project
C. Chambal Project
D. Kosi Project

Q64. The Tehri Dam, which serves multiple purposes including power generation, is built on which river?

RRB NTPC UG CBT-01 2024 (13 Aug, 2025 Shift 1)

A. Yamuna
B. Ganga
C. Sutlej
D. Bhagirathi

Q65. Barapani Lake of Shillong is also known as _____.

RRB NTPC UG CBT-01 2024 (08 Aug, 2025 Shift 3)

A. Umiam Lake
B. Shilloi Lake
C. Haflong Lake
D. Jaisamand Lake

- **Largest Artificial Lake in India:** Govind Ballabh Pant Sagar (on Rihand river, UP). Named after UP's first CM.
- **Second Largest Artificial Lake:** Dhebar Lake (Jaisamand Lake) in Udaipur, Rajasthan.

Q66. Which is the largest saline wetland in India?

RRB NTPC Graduation CBT-01 2024 (20 Jun, 2025 Shift 2)

A. Wular Lake
B. Pulicat Lake
C. Chilika Lake
D. Sambhar Lake

- **If "inland" was asked, it would be Sambhar. But overall largest saline wetland is Chilika. Pulicat is the second largest.**

Q67. Which of the following lakes in Rajasthan is artificial?

RRB NTPC Graduation CBT-01 2024 (17 Jun, 2025 Shift 1)

A. Ramgarh
B. Ana Sagar
C. Pushkar
D. Sambhar

- **Located in Ajmer, built by Arnoraja (grandfather of Prithviraj Chauhan) between 1135-1150 AD.**



National Parks & Wildlife Sancutaries

Q68. The Pitti Island in Lakshadweep is notable for which feature?

NTPC UG CBT-01 2025-26 (19 Jun, 2026 (9:00 AM - 10:30 AM)

A. It hosts India's largest military base
B. It is the gateway to the Andaman Sea
C. It contains active volcanoes
D. It is uninhabited and has a bird sanctuary

Q69. Rajasthan's Desert National Park (Desert National Park) is located in which districts of the state?

RRB NTPC UG CBT-01 2024 (09 Sept, 2025 Shift 1)

A. Jaisalmer and Barmer
B. Phalodi and Jodhpur
C. Udaipur and Rajsamand
D. Dhaulpur and Bharatpur

Q70. In which of the following Indian states is the Sundarbans Biosphere Reserve located?

A. Tamil Nadu

RRB NTPC UG CBT-01 2024 (14 Aug, 2025 Shift 2)

B. West Bengal
C. Assam
D. Odisha

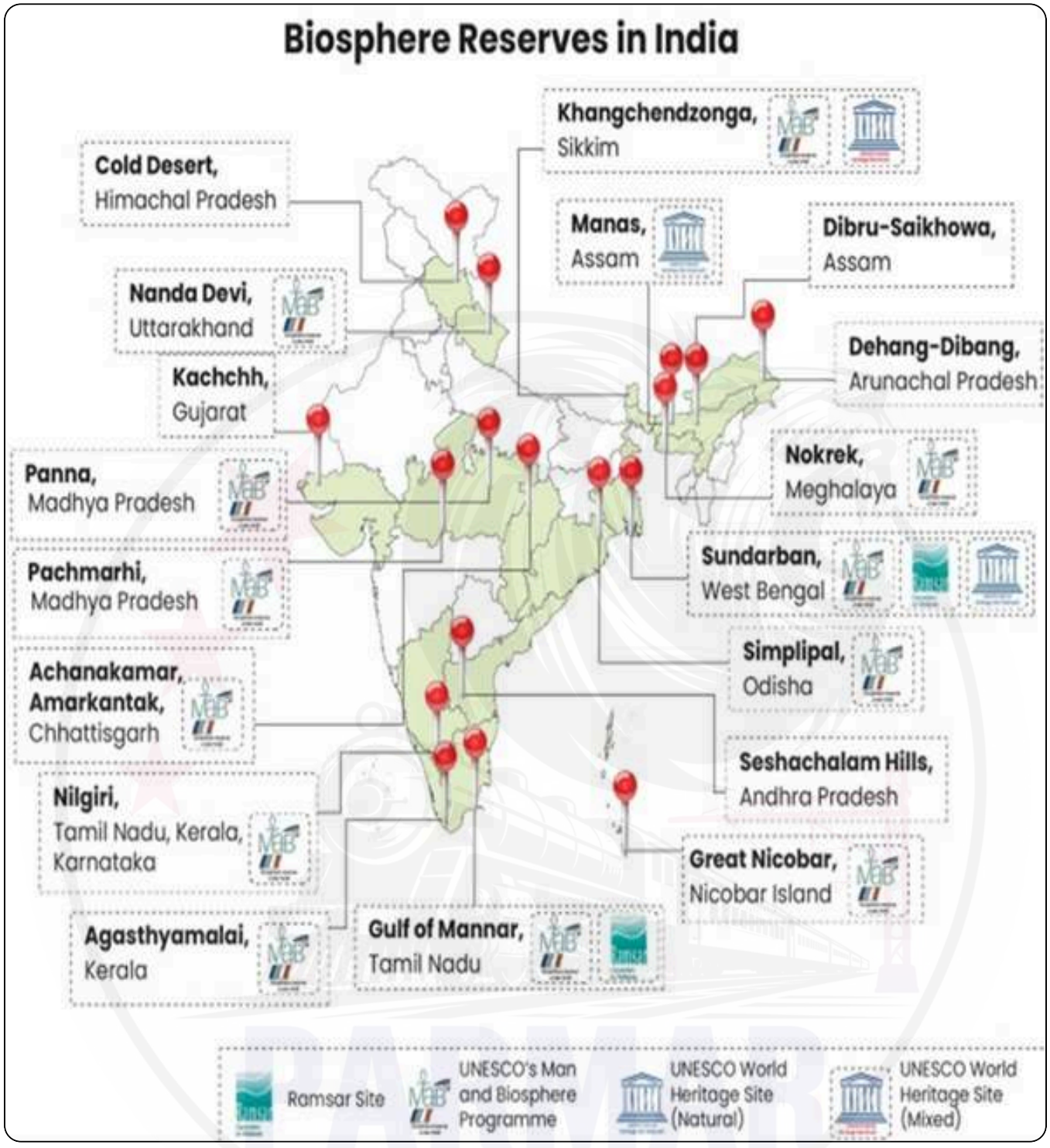
Q71. Which of the following is the first biosphere reserve established in India?

A. Nilgiri

RRB NTPC UG CBT-01 2024 (07 Aug, 2025 Shift 2)

B. Sundarbans
C. Nanda Devi
D. Gulf of Mannar

- **Biosphere Reserves in India: 18 total. 13 are recognized under UNESCO's MAB (Man and the Biosphere) program.**
  - **Nilgiri**: 1st Biosphere Reserve (Recognized by UNESCO in 2000). Spans Tamil Nadu, Kerala, Karnataka.
  - **Gulf of Mannar:** Tamil Nadu.
  - **Sundarbans**: West Bengal (Famous for Royal Bengal Tiger).
  - **Nanda Devi**: Uttarakhand.
  - **Nokrek**: Meghalaya.
  - **Pachmarhi**: MP (Queen of Satpura).
  - **Simlipal**: Odisha.
  - **Achanakmar-Amarkantak**: MP and Chhattisgarh border.
  - **Great Nicobar**: A&N Islands.
  - **Agasthyamalai**: (Kerela & Tamil Nadu) - Recognized in 2016.
  - **Khangchendzonga (Kanchenjunga):** Sikkim (UNESCO's only Mixed World Heritage Site, 2018).
  - **Panna**: MP.
  - **The Cold Desert Biosphere Reserve in Himachal Pradesh** is India's 13th site recognized under UNESCO's Man and the Biosphere (MAB) Programme.



Q72. Which of the following is a National Park in Rajasthan?

RRB NTPC Graduation CBT-01 2024 (17 Jun, 2025 Shift 2)

A. Kaziranga National Park
B. Ranthambore National Park
C. Sundarbans National Park
D. Jim Corbett National Park

- **Assam**: Kaziranga (One-horned rhino), Orang, Manas, Dibru-Saikhowa.
- **Rajasthan**: Ranthambore, Desert NP.
- **Uttarakhand**: Jim Corbett (1st NP in India, 1936, formerly Hailey NP, Project Tiger launched here in 1973), Valley of Flowers, Nanda Devi, Rajaji NP.
- **West Bengal:** Buxa, Neora Valley.

- **Madhya Pradesh (Most NPs in India)**: Bandhavgarh, Madhav, Kanha, Panna, Kuno (Cheetahs introduced here), Pench (Border of MP/MH, night safari opened).
- **Uttar Pradesh:** Dudhwa NP.
- **Gujarat**: Gir (Asiatic Lions), Blackbuck, Vansda.

Q73. As of April 2025, Similipal – designated as the second national park of the state – is located in:

RRB NTPC Graduation CBT-01 2024 (09 Jun, 2025 Shift 2)

A. Gujarat
B. Odisha
C. Madhya Pradesh
D. Assam

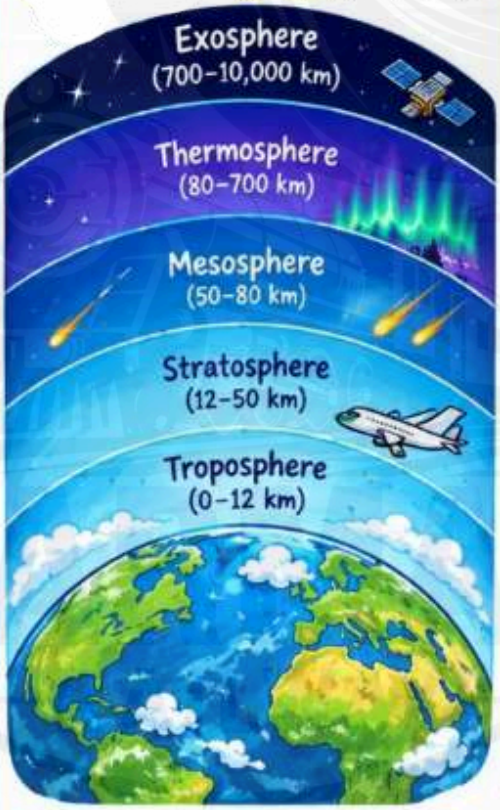
- **Bhitarkanika is the first NP of Odisha; Simlipal is the second.**

Atmosphere

Layers of Atmosphere

5 Exosphere
- **Altitude** : 700 – 10,000 km (gradual transition to outer space)
- **Temperature** : Very high (> 1000°C)
- **Features** : Extremely thin air, mainly hydrogen and helium.
- **Importance** : Satellites orbit in this layer.

4 Thermosphere
- **Altitude** : 80 – 700 km
- **Temperature** : Very high (500°C – 2000°C)
- **Features** : Absorbs high-energy solar radiation (UV and X-rays); air is very thin.
- **Importance** : Ionosphere is present in this layer (aids radio communication). Auroras (Northern and Southern Lights) occur here.



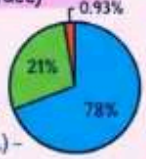
3 Mesosphere
- **Altitude** : 50 – 80 km
- **Temperature** : Decreases with height (–90°C, coldest layer)
- **Features** : Air is thin and density is low.
- **Importance** : Meteors burn up in this layer due to friction.

2 Stratosphere
- **Altitude** : 12 – 50 km
- **Temperature** : Increases with height (–60°C to 0°C)
- **Features** : Contains ozone layer.
- **Importance** : Ozone layer absorbs harmful UV radiation from the Sun.



1 Troposphere
- **Altitude** : 0 – 12 km (average ~12 km)
- **Temperature** : Decreases with height (15°C to –56°C)
- **Features** : Densest layer; contains 75% of total air mass and almost all water vapour.
- **Importance** : Weather phenomena (clouds, rain, snow, winds) occur in this layer.

Composition of Atmospheric Air (Near Earth's Surface)
- Nitrogen (N_2) – 78%
- Oxygen (O_2) – 21%
- Argon (Ar) – 0.93%
- Carbon dioxide (CO_2) – 0.04%
- Other gases (Ne, He, CH_4, Kr, etc.) – traces



Key Points
- **Temperature trend** : Decreases in troposphere and mesosphere, increases in stratosphere and thermosphere.
- **Density of air** : Maximum at earth's surface, decreases with altitude.
- **Weather** : Occurs only in troposphere.
- **Ozone layer** : Present in stratosphere.
- **Satellites** : Orbit in exosphere (and low thermosphere).

Mnemonics (Order from Surface to Space)
- T → Troposphere (0 – 12 km)
- S → Stratosphere (12 – 50 km)
- M → Mesosphere (50 – 80 km)
- T → Thermosphere (80 – 700 km)
- E → Exosphere (700 – 10,000 km)

Mnemonics : "Today Students Must Take Exams"

- **Layers of the Atmosphere**
  - **Troposphere** (Kshobh Mandal): Lowest layer. Average height 13 km (18 km at equator due to convection currents lifting heat, 8 km at poles). All weather phenomena occur here. Temperature decreases with height.
    - This decrease is called the **Lapse Rate** (drops by 6.5°C per 1 km, or 1°C per 165 meters). Ends at Tropopause.
  - **Stratosphere**: Extends up to 50 km. Temperature increases with height. Contains the Ozone Layer which protects from harmful UV rays.
    - Airplanes fly here to avoid weather turbulence and clouds.
    - Ends at Stratopause. (**Ozone Day:** Sept 16, Montreal Protocol 1987).
  - **Mesosphere**: Extends up to 85 km. Temperature decreases with height. It is the coldest layer. Meteorites burn up here.
  - **Ionosphere / Thermosphere:** Temperature increases rapidly. Contains ions that reflect Radio Waves back to earth.
  - **Exosphere**: Outermost layer.

- **Gaseous Composition in Troposphere:** Nitrogen (78%), Oxygen (21%), Argon (0.9%), CO2 (0.036%).



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